WELLSENSE · DESIGN DECISIONS REGISTER

WellSense design decisions register

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	WLS-DEC-001	Version	0.5
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/wellsense		
Source	docs/06-design-decisions.md @ d54201d		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-25	Amish Chadha	Amish kept D-S1 to D-S4 (O-2) and accepted the installed render layout, 2026-10-25 ("i concur with your recommendations"); O-1 stays open (A and C within 10 %, no pick)
0.4	2026-10-25	Amish Chadha	Kit 1.9.1 upgrade; concept selection between options A and C opened as O-1 (proposed, awaiting Amish, WLS-DDR-005); fixes D-S1 (clevis pin on the tube collar) and D-S2 (drift test of the tube string) decided under Amish's standing instruction of 2026-10-23; value-engineering cost restated at USD 263.90
0.3	2026-10-02	Amish Chadha	Locks modelled and priced (BOM line 16); value-engineering cost restated at USD 262.40
0.2	2026-10-02	Amish Chadha	Amish approved the recommendations for open items 1 to 9 on 2026-10-02 (WLS-DDR-004 accepted with A1 and A2, ACWADAM as first candidate partner, standard FieldNode core and pinout, R15 kept

			with hot sites held, render items); moved to decisions made
0.1	2026-10-02	Amish Chadha	Register opened with the build plan; budget treated as a value-engineering target

WellSense design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

Open decisions

Table 1. Decisions awaiting Amish.

#	Decision	Options	Recommendation	Record
O-1	Concept to carry forward	A, vented transducer in an access tube (current design, 173 of 215); C, acoustic echo sounder (164); B, bubbler (147, not put forward)	None: A and C are within 10 %, so both go to Amish without a pick (STANDARDS section 21). Proposed, awaiting Amish; the repo stays on A meanwhile	WLS-OPT-001, WLS-DDR-005

Decided under Amish's standing instruction of 2026-10-23 (fix rule)

Amish, 2026-10-23: when a STEM check row fails, propose and implement the best fix for the design and its constructability in the same session and record it here as decided under that instruction. Amish can reverse either.

Table 2. Fixes made on 2026-10-25 (D-S3 and D-S4 after the independent review of WLS-STM-001).

#	STEM row	Fix	Options weighed	Cost	Implemented in
D-S1	S14: the split collar's friction grip on PVC (833 N new, about 278 N after creep) is short of 3 x 264 N, and losing it drops the tube string into the well	A 5 mm stainless clevis pin (ISO 2341 B) with a split pin through the access tube on a chord 9 mm off the axis, 2 mm above the collar; the collar keeps the grip and the pin takes the load if it slips (PVC bearing 7.8 MPa, pin shear 6.7 MPa)	(a) clevis pin, chosen: cheapest, a drill only, clear of the cable; (b) solvent-welded shoulder coupling at the plate: needs the string cut on site and a larger hole in the split plate; (c) a second collar: same creep	+USD 1.50 (estimate), BOM line 4	`cad/src/model.py`, BOM line 4, WLS-CAL-001 [D8], WLS-BLD-001 section 3.8 and step 14, PR-002
D-S2	S18: a 22 mm probe has 1.74 mm radial clearance at the thickest Sch 40 wall and 1.24 mm at a cement bead, under the 2 mm rule	Wipe every solvent-weld joint inside while wet and pull a 23 mm drift (a 300 mm dowel on a cord) through the whole string before the probe goes in; a 19 mm probe stays acceptable under the "22 mm or less" specification	(a) wipe and drift, chosen: no cost, catches any snag before the probe is at risk; (b) specify a 19 mm probe: narrows suppliers, price unknown; (c) a larger tube: does not pass a centered riser in 150 mm casing	None	WLS-CAL-001 [D9], WLS-BLD-001 section 3.8 and step 12, PR-010
D-S3	S7 (found by the independent review): a span calibration from a 1 m lift rests on two tape readings (±3 mm) and two probe repeatability terms (±10 mm), about 1.5 % of span, 148 mm at 10 m	Offset from the installation tape reading; span from the probe's per-unit factory calibration certificate (zero and span), refined by quarterly tape checks once two are 3 m or more apart; the 1 m lift stays as a response check	(a) certificate span, chosen: most makers of 0.25 % class probes supply one; (b) a longer lift: a 5 m service loop does not fit the box and still leaves about 0.3 %; (c) datasheet span only: 25 mm at full head	None priced; the certificate's cost is to confirm with the supplier (BOM line 1)	WLS-CAL-001 [C3], [C4], [C4b], PRC step 7, PR-004, BOM line 1, BLD first checks
D-S4	S28 (found by the independent review): a short between the cable cores puts the 24 V boost across the shunt, 149 mA and 3.3 W	A 50 mA resettable fuse (PTC) in series with the loop on the interface board; normal 22 mA is below its hold current	(a) PTC fuse, chosen: resets itself, a few cents; (b) series current-limit resistor: costs loop compliance (20.16 V would fall towards 12 V); (c) rely on the boost module's current limit: not specified for low-cost modules	+USD 0.50 (estimate), BOM line 7	WLS-CAL-001 [B1c], BOM line 7, PR-018

To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	Drinking-water certificates for the probe, the vented cable, the HDPE sheet and the EPDM (gasket and wraps)	R9 is not met without them	WLS-REQ-001, R9; WLS-DDR-004
2	The riser and pump cable diameters at the well	They set the seal plate's split holes (designed for a 42.2 mm riser and a 12 mm cable)	WLS-DDR-004, P6
3	The junction box model, its boss spacing (80 x 110 mm assumed) and its lug kit	They set the internal plate holes and the lug holes in the plate	WLS-DDR-004, P3
4	The shaft collar's rated axial grip on PVC pipe, at least 264 N	It carries the tube at 60 m; the clevis pin of D-S1 is the backup if it creeps	WLS-CAL-001, [D5], [D8]
5	The cable support grip suits 6 to 8 mm cable and carries at least 35 N	It carries the probe and cable at 60 m	WLS-CAL-001, [D6]
6	The probe body is 22 mm or less, and its supplier's price	2.3 mm radial clearance in the tube; the BOM assumes the 26 to 28 mm class price	WLS-DDR-002; BOM line 1
7	The flexible tail's connector fits a 22.5 mm hole in a 1 in PVC cap, with its locknut inside	The cap is the connector's only support	WLS-DDR-004, P5 and P8

Value engineering

Value-engineering target: USD 200 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 264.40 for the WellSense parts at the 30 m design depth (USD 64.40 over the target); USD 403.40 with the FieldNode core, which is costed in the FieldNode repo. The clevis pin of D-S1 added USD 1.50 and the fuse of D-S4 USD 0.50 on 2026-10-25. Main cost drivers and savings worth trying:

- The largest lines are the transducer (USD 50), the access tube (USD 31 at 30 m), the post with its plate, V-blocks and bands (USD 32), the vented cable (USD 26.40 at 30 m), the seal plate set (USD 24.50 with the clevis pin) and the conduit set (USD 20). Cable and tube scale with depth at USD 1.80 a metre.
- Making the design constructable added USD 46.80: the seal plate set (+8), tube cap and grip (+4), junction box lugs, glands and internal plate (+6), terminal strip (+1), junction box plate and V-blocks (+10), flexible tail, saddles and hub (+10), the FieldNode lead (+7) and 1 m more cable (+0.80).
- The locks decided on 2026-10-02 added USD 18.00: line 16 (security-head screw and bit USD 3.50, two padlocks USD 9.00, hasp kit USD 4.00) and USD 1.50 for the eye bolt in line 5. The lockable steel wellhead cover is priced only when a site visit calls for it.
- Savings worth trying: fit the conduit set only where livestock or tampering is likely (USD 20); leave the access tube out where no pump shares the casing (USD 31 at 30 m, already the rule); mount the barometric module inside the junction box, which breathes through its desiccant, instead of in its own housing (about USD 4); buy the junction box with a pole-mount kit if one is offered at less than the plate, V-blocks and bands (up to USD 10); cut the seal plate from offcuts shared across several wells.

Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 items D1 to D8: FieldNode core costed in its own repo; vented 4 to 20 mA transducer; 0 to 10 m range; own access tube where a pump shares the casing; 15 min readings, 1 min in pumping tests; quarterly manual tape check; levels only, community controls sharing; FieldNode as the core	Amish: "i accept all your recommendations, go with them across all repos."	WLS-DDR-001, WLS-DDR-002
2026-09-25	TRL 3 items A9 to A13: design depth 30 m; 1 in tube with a probe of 22 mm or less; galvanized conduit over the surface cable; FieldNode sun shield at hot sites; pumping tests send every 30 min on a public network	Amish, same instruction	WLS-DDR-002
2026-09-26	`budget_usd` set to $200	Amish: "i approve all the budget items."	WLS-DDR-002, N1
2026-09-27	FieldNode panel tilt corrected to face the way the node faces	Amish: "resolve the challenges for ConePro, BridgePulse, Grainguard and WellSense."	WLS-DDR-003
2026-09-30	Make the design physically buildable while drawing the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations." Changes made under this instruction; open for his review (open decision 1)	WLS-DDR-004
2026-10-01	`budget_usd` is a value-engineering target, not a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	STANDARDS section 18; this register
2026-10-02	Design for construction accepted: the changes P1 to P11 of WLS-DDR-004, as made	Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."	WLS-DDR-004
2026-10-02	Locking (R16): option (a) is fitted on the prototype from the start: a security-head screw on the rim band, a padlock through the cross bolt's head and a padlockable hasp kit on the junction box; the lockable steel wellhead cover is used at sites where the first site visit shows open access or livestock	Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."	WLS-DDR-004, A1; WLS-PRC-001 open questions
2026-10-02			WLS-DDR-001, O1

Date	Decision	Decided by	Record
	First partner and region: a participatory groundwater management group in India that works with farmer groups on shared aquifers; the first candidate to approach is ACWADAM in Pune, with its well sizes setting the design well and the FieldNode core at its sites on the IN865 band	Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."	
2026-10-02	Battery-only FieldNode variant: the standard FieldNode core with its panel is kept; no battery-only variant is asked of FieldNode until a site shows the panel cannot be placed	Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."	WLS-DDR-001, O2
2026-10-02	FieldNode sensor port pinout: FieldNode's proposed standard pinout is signed off for port A (pin 1 switched rail, pin 2 data A, pin 3 ground, pin 4 data B, pin 5 analog), with data A and B carrying I2C; the switched rail voltage is confirmed against the boost module input	Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."	FND-DDR-001, O2; FND-DEC-001
2026-10-02	FieldNode core above 45 °C: R15 stays at 55 °C, as decided on 2026-09-25 (WLS-DDR-002, A12); the FieldNode project is asked to confirm the shielded core at 55 °C ambient, and until it does WellSense is not sited where the design maximum exceeds 45 °C	Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."	REVIEW.md, cross-repo actions; WLS-DDR-002, A12
2026-10-02	FieldNode geometry in `cad/vendor/`: kept as a copy and re-exported whenever FieldNode's general arrangement is revised, as a line in the review checklist	Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."	WLS-DDR-004, A2
2026-10-02	Photoreal render layout: the post drawn closer to the well and lower than installed is accepted, for renders only	Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."	REVIEW.md, 2026-09-26, render item 1
2026-10-02	Exploded render: grouping the post, tube and cable together is accepted	Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."	REVIEW.md, 2026-09-26, render item 5
2026-10-25	Fixes D-S1 (clevis pin on the tube collar), D-S2 (drift test of the tube string), D-S3 (span from the probe's factory certificate) and D-S4 (loop fuse), from STEM rows S14, S18, S7 and S28	Decided by Claude under Amish's standing instruction of 2026-10-23 (fix rule); kept by Amish on 2026-10-25 (O-2)	WLS-STM-001; this register, Table 2

Date	Decision	Decided by	Record
2026-10-25	O-2: keep all four fixes D-S1 to D-S4 (clevis pin, drift test, span from the factory certificate, loop fuse), as made	Amish: "i concur with your recommendations"	WLS-STM-001; this register, Table 2
2026-10-25	Product render layout: the logger drawn at its installed layout (post 0.75 m from the well, FieldNode at 1.75 m), every part from `components(CHOSEN)`, in place of the closer, lower render layout accepted on 2026-10-02 (as proposed on 2026-10-24)	Amish: "i concur with your recommendations"	`cad/src/product_model.py` (no change); `docs/REVIEW.md`, session 2026-10-24